

2024-2025 Fall
MAT123 Final
09/01/2025

1. Evaluate the following integrals.

a. $\int \frac{dx}{2 \cos x + 3}$

b. $\int \frac{\sqrt{5-x^2}}{x} dx$

c. $\int \frac{dx}{1-x^4}$

d. $\int \ln(1+x^2) dx$

2. Consider the region R bounded by the curves $y = x$, $y = x^2$ and $y = x^2/2$.

a. Find the area of the region R .

b. Write a definite integral (do not evaluate) by using the Washer Method, which gives the volume of a solid obtained by rotating the region R about the y -axis.

c. Write a definite integral (do not evaluate) by using the Cylindrical Shell Method, which gives the volume of the solid obtained by rotating the region R about the line $y = 2$.

3. Use the Integral Test to determine the existence of the sum of the series $\sum_{n=1}^{\infty} ne^{-n^2}$.

4. Determine whether the series $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{n}\right)^{1/n}$ is convergent or divergent.

5. Find the Maclaurin series of $f(x) = \sqrt{e^x}$ and determine the interval of convergence of this series.

1. a. Use the tangent half-angle substitution, which is also called the Weierstrass substitution. Let $t = \tan\left(\frac{x}{2}\right)$. Then

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{2}{1+t^2} dt$$

$$\int \frac{dx}{2\cos x + 3} = \int \frac{1}{2\left(\frac{1-t^2}{1+t^2}\right) + 3} \cdot \frac{2}{1+t^2} dt = \int \frac{2}{5+t^2} dt = \frac{2}{5} \int \frac{1}{1+\left(\frac{t}{\sqrt{5}}\right)^2} dt$$

Let $u = \frac{t}{\sqrt{5}}$, then $\sqrt{5} du = dt$.

$$\frac{2}{5} \int \frac{1}{1+\left(\frac{t}{\sqrt{5}}\right)^2} dt = \frac{2\sqrt{5}}{5} \int \frac{1}{1+u^2} du$$

$$= \frac{2\sqrt{5}}{5} \arctan u + C = \frac{2\sqrt{5}}{5} \arctan \frac{t}{\sqrt{5}} + C$$

$$= \boxed{\frac{2\sqrt{5}}{5} \arctan \left(\frac{1}{\sqrt{5}} \cdot \tan \left(\frac{x}{2} \right) \right) + C, \quad C \in \mathbb{R}}$$

- b. Let $x = \sqrt{5} \sin u$ for $-\frac{\pi}{2} \leq u < \frac{\pi}{2}$, then $dx = \sqrt{5} \cos u du$.

$$I = \int \frac{\sqrt{5-x^2}}{x} dx = \int \frac{\sqrt{5-5\sin^2 u}}{\sqrt{5} \sin u} \cdot \sqrt{5} \cos u du \quad [\sin^2 u + \cos^2 u = 1]$$

$$= \sqrt{5} \int \sqrt{\cos^2 u} \cot u du \quad [\cos u > 0]$$

$$= \sqrt{5} \int \cos u \cdot \cot u du = \sqrt{5} \int \frac{\cos^2 u}{\sin u} du = \sqrt{5} \int \frac{1 - \sin^2 u}{\sin u} du$$

$$= \sqrt{5} \int (\csc u - \sin u) du = \sqrt{5} (-\ln |\cot u + \csc u| + \cos u) + C$$

Recall that $x = \sqrt{5} \sin u$.

$$\sin u = \frac{x}{\sqrt{5}} \implies \sin^2 u = \frac{x^2}{5} \implies \cos^2 u = \frac{5-x^2}{5} \implies \cos u = \frac{\sqrt{5-x^2}}{\sqrt{5}}$$

$$\cot u = \frac{\cos u}{\sin u} = \frac{\sqrt{5-x^2}}{x}, \quad \csc u = \frac{1}{\sin u} = \frac{\sqrt{5}}{x}$$

Therefore,

$$I = -\sqrt{5} \ln \left| \frac{\sqrt{5-x^2}}{x} + \frac{\sqrt{5}}{x} \right| + \sqrt{5-x^2} + C, \quad C \in \mathbb{R}$$

c. Use partial fractions to compute the integral.

$$\begin{aligned} I &= \int \frac{1}{1-x^4} dx = \int \frac{1}{(1-x^2)(1+x^2)} dx = \int \frac{1}{(1-x)(1+x)(1+x^2)} dx \\ &= \int \left(\frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2} \right) dx \end{aligned}$$

$$\begin{aligned} A(1+x)(1+x^2) + B(1-x)(1+x^2) + (Cx+D)(1+x)(1-x) &= 1 \\ x^3(A-B-C) + x^2(A+B-D) + x(A-B+D) + A+B+D &= 1 \end{aligned}$$

Equate the coefficients of like terms.

$$A - B - C = 0 \tag{1}$$

$$A + B - D = 0 \tag{2}$$

$$A - B + C = 0 \tag{3}$$

$$A + B + D = 1 \tag{4}$$

Solve for A , B , C , and D .

$$(1) \& (3) \rightarrow 2A - 2B = 0$$

$$(2) \& (4) \rightarrow 2A + 2B = 1$$

$$\therefore A = \frac{1}{4}, \quad B = \frac{1}{4}$$

$$(1) \rightarrow C = 0$$

$$(2) \rightarrow D = \frac{1}{2}$$

Rewrite the integral.

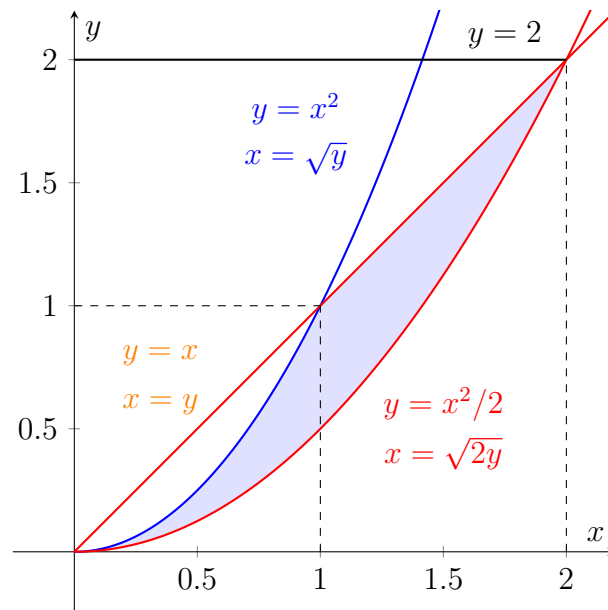
$$\begin{aligned} I &= \int \left(\frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2} \right) dx \\ &= \int \left(\frac{1}{4(1-x)} + \frac{1}{4(1+x)} + \frac{1}{2(1+x^2)} \right) dx \\ &= \left[-\frac{1}{4} \ln |1-x| + \frac{1}{4} \ln |1+x| + \frac{1}{2} \arctan(x) + C_0, \quad C_0 \in \mathbb{R} \right] \end{aligned}$$

d. Use the method of integration by parts.

$$\left. \begin{array}{l} u = \ln(1+x^2) \implies du = \frac{2x}{1+x^2} dx \\ dv = dx \implies v = x \end{array} \right\} \rightarrow \int u dv = uv - \int v du$$

$$\begin{aligned} \int \ln(1+x^2) dx &= x \ln(1+x^2) - \int \frac{2x^2}{1+x^2} dx \\ &= x \ln(1+x^2) - \int \frac{2x^2 + 2 - 2}{1+x^2} dx \\ &= x \ln(1+x^2) - 2 \int dx + 2 \int \frac{dx}{1+x^2} \\ &= \boxed{x \ln(1+x^2) - 2x + 2 \arctan x + C, \quad C \in \mathbb{R}} \end{aligned}$$

2.



$$\begin{aligned} \text{a.} \quad A &= \int_0^1 \left(x^2 - \frac{x^2}{2} \right) dx + \int_1^2 \left(x - \frac{x^2}{2} \right) dx = \left[\frac{x^3}{3} - \frac{x^3}{6} \right]_0^1 + \left[\frac{x^2}{2} - \frac{x^3}{6} \right]_1^2 \\ &= \left[\left(\frac{1}{3} - \frac{1}{6} \right) - 0 \right] + \left[\left(2 - \frac{8}{6} \right) - \left(\frac{1}{2} - \frac{1}{6} \right) \right] = \boxed{\frac{1}{2}} \end{aligned}$$

$$\text{b.} \quad V = \int_D \pi [r_2^2(y) - r_1^2(y)] dy$$

$$\boxed{V = \int_0^1 \pi \left[(\sqrt{2y})^2 - (\sqrt{y})^2 \right] dy + \int_1^2 \pi \left[(\sqrt{2y})^2 - y^2 \right] dy}$$

c. $V = \int_D 2\pi \cdot h(y) \cdot r(y) dy$

$$V = \int_0^1 2\pi(2-y) \left(\sqrt{2y} - \sqrt{y} \right) dy + \int_1^2 2\pi(2-y) \left(\sqrt{2y} - y \right) dy$$

3. Take $f(x) = xe^{-x^2}$. f is continuous because the product of a polynomial and an exponential expression is still continuous. f is positive and decreasing for $x \geq 1$. Verify the monotonicity of f by taking the first derivative.

$$\frac{df}{dx} = 1 \cdot e^{-x^2} + xe^{-x^2} \cdot (-2x) = e^{-x^2} (1 - 2x^2)$$

$$f'(x) < 0 \quad \text{for } x > \frac{\sqrt{2}}{2} \implies f'(x) < 0 \quad \text{for } x \geq 1$$

We may now apply the Integral Test. Take the limit for the improper integral.

$$\int_1^\infty xe^{-x^2} dx = \lim_{R \rightarrow \infty} \int_1^R xe^{-x^2} dx = \lim_{R \rightarrow \infty} -\frac{1}{2}e^{-x^2} \Big|_1^R = \lim_{R \rightarrow \infty} -\frac{1}{2} \left(e^{-R^2} - e^{-1} \right) = \frac{1}{2e}$$

The integral converges. Then the series $\sum_{n=1}^\infty ne^{-n^2}$ also converges.

4. Apply the n th Term Test for the non-alternating part. Let L be the value of the limit.

$$L = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)^{1/n} \implies \ln(L) = \ln \left[\lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)^{1/n} \right]$$

We can also take the limit inside the function because $\ln x$ is continuous on its domain.

$$\ln(L) = \lim_{n \rightarrow \infty} \ln \left[\left(\frac{1}{n} \right)^{1/n} \right] = \lim_{n \rightarrow \infty} \frac{\ln \left(\frac{1}{n} \right)}{n}$$

To be able to use L'Hôpital's rule, take the corresponding function $f(x) = \frac{\ln \left(\frac{1}{x} \right)}{x}$.

$$\lim_{x \rightarrow \infty} \frac{\ln \left(\frac{1}{x} \right)}{x} \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \infty} \frac{x \cdot \left(-\frac{1}{x^2} \right)}{1} = \lim_{x \rightarrow \infty} -\frac{1}{x} = 0 \implies \ln(L) = 0 \implies L = 1$$

The limit $\lim_{n \rightarrow \infty} (-1)^{n+1}$ does not exist because of oscillation. So, $\lim_{n \rightarrow \infty} (-1)^{n+1} \left(\frac{1}{n} \right)^{1/n}$ does not exist either. Therefore, by the n th Term Test for divergence, the series $\sum_{n=1}^\infty (-1)^{n+1} \left(\frac{1}{n} \right)^{1/n}$ is divergent.

5. The Maclaurin series of f is given by $\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$. Find $f(0)$, $f'(0)$, $f''(0)$, $f'''(0)$ to look for the pattern.

$$f'(x) = \frac{1}{2\sqrt{e^x}} \cdot e^x = \frac{1}{2}\sqrt{e^x}, \quad f''(x) = \frac{1}{4\sqrt{e^x}} \cdot e^x = \frac{1}{4}\sqrt{e^x}, \quad f'''(x) = \frac{1}{8\sqrt{e^x}} \cdot e^x = \frac{1}{8}\sqrt{e^x}$$

$$f(0) = 1, \quad f'(0) = \frac{1}{2}, \quad f''(0) = \frac{1}{4}, \quad f'''(0) = \frac{1}{8}$$

Therefore, $f^{(k)}(0) = \left(\frac{1}{2}\right)^k$. Rewrite the summation formula.

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = \sum_{k=0}^{\infty} \frac{\left(\frac{1}{2}\right)^k \cdot x^k}{k!} = \boxed{\sum_{k=0}^{\infty} \frac{x^k}{2^k \cdot k!} = 1 + \frac{x}{2} + \frac{x^2}{8} + \frac{x^3}{48} + \cdots}$$

Find the interval of convergence by applying the Ratio Test.

$$\lim_{k \rightarrow \infty} \left| \frac{x^{k+1}}{2^{k+1} \cdot (k+1)!} \cdot \frac{2^k \cdot k!}{x^k} \right| = \lim_{k \rightarrow \infty} \frac{x}{2k} = 0 < 1$$

The series is convergent on $\mathbb{R}$.